

Numerical Physics I — Exercise 1

Modelisation 0D of runaway electrons. Euler schemes: explicit, implicit and semi-implicit

Pasqual Hernández Riveiro
pasqual.hernandezriveiro@epfl.ch

Jordi Mitjà Cortés
jordi.mitjacortes@epfl.ch

March 10, 2026

Contents

1	Introduction	2
2	Analytical calculations	2
2.1	Theory and data	2
2.2	Steady-state solution	2
2.3	Exact time-dependent solution	3
2.4	Limits and asymptotics behaviors	4
3	Numerical method and implementation	5
3.1	Unified Euler scheme	5
3.2	Time grid and outputs	5
3.3	Iterative treatment of the implicit and semi-implicit schemes	5
4	Simulations and analyses	6
4.1	(a) Case $\bar{\gamma} > 0$: convergence of $\bar{n}(\bar{t}_{\text{final}})$ and $\bar{\tau}$	6
4.2	(b) Early-time behaviour: $\beta\bar{t} \ll 1$	8
4.3	(c) Case $\bar{\gamma} < 0$: loss-dominated regime	9
4.4	(d) Stability study (explicit Euler)	9
4.5	(e) (Semi-)implicit scheme without iteration (engine2.cpp)	11
4.6	(f) Optional: efficiency for a target accuracy	12
5	Conclusion	13

1 Introduction

Runaway electrons may appear in a plasma when the accelerating effect of the electric field overcomes collisional friction. In that regime, energetic electrons can be continuously accelerated and may generate additional runaway electrons through avalanche multiplication. Their temporal evolution results from the competition between external sources, avalanche growth, transport losses and nonlinear saturation effects.

In this work, we study a simplified zero-dimensional model for the runaway electron density. The problem is first analysed analytically in order to derive the steady-state solution, the exact time-dependent solution, and several useful asymptotic limits. The model is then solved numerically using explicit, implicit and semi-implicit Euler schemes.

The numerical study focuses on the convergence properties of the schemes, the early-time behaviour in the regime $\beta\bar{t} \ll 1$, and the comparison between avalanche-dominated and loss-dominated regimes. Additional investigations are carried out on the stability of the explicit method and on alternative implementations of the implicit and semi-implicit updates.

2 Analytical calculations

2.1 Theory and data

When the electric field in a plasma exceeds the Connor–Hastie critical field, energetic electrons can continuously accelerate and generate additional runaway electrons through avalanche multiplication. The evolution of the runaway electron density $n(t)$ can be modeled by [Decker NF 2024]:

$$\frac{dn}{dt} = d + \left[\gamma \left(1 - \frac{n}{n_R} \right) - \nu \right] n \quad (1)$$

Here, d represents an external source of runaway electrons density (e.g. Dreicer generation or tritium beta decay), γ is the avalanche growth rate, n_R is the runaway electron density required to carry the total toroidal plasma current, and ν represents a loss rate generally due to radial transport.

In order to normalize the equation, we introduce the dimensionless variables:

$$\bar{n} = \frac{n}{n_R}, \quad \bar{t} = \gamma t, \quad \bar{d} = \frac{d}{\gamma n_R}, \quad \bar{\nu} = \frac{\nu}{\gamma} \quad \bar{\gamma} = 1 - \bar{\nu}.$$

Thus, the normalized equation becomes

$$\frac{d\bar{n}}{d\bar{t}} = \bar{d} + (\bar{\gamma} - \bar{n})\bar{n} \quad (2)$$

This nonlinear differential equation will be solved numerically and analytically in the following sections.

2.2 Steady-state solution

Starting from (2), in order to find the steady state solution we can just impose that $\frac{d\bar{n}}{d\bar{t}} \rightarrow 0$ due to the stationary nature of this state. That leaves equation (2) as

$$\bar{d} + (\bar{\gamma} - \bar{n})\bar{n} = 0 \quad (3)$$

Solving the equation for $\bar{n}$ yields two possible solutions. We retain the positive root since it is the only one with physical meaning. Because $\bar{d}$ is always positive, choosing the negative root would lead to a negative steady-state density, which is not physically meaningful.

$$\bar{n}_\infty = \frac{\bar{\gamma} + \sqrt{\bar{\gamma}^2 + 4\bar{d}}}{2} \quad (4)$$

For the case where $\bar{d} \ll \bar{\gamma}^2$, solution (4) simplifies into a more compact expression. Knowing that if $\bar{d} \ll \bar{\gamma}^2$

$$\sqrt{\bar{\gamma}^2 + 4\bar{d}} = \bar{\gamma} \sqrt{1 + \frac{4\bar{d}}{\bar{\gamma}^2}} \sim \bar{\gamma} \left(1 + \frac{2\bar{d}}{\bar{\gamma}^2}\right)$$

Then

$$\bar{n}_\infty \sim \frac{1}{2} \left(\bar{\gamma} + |\bar{\gamma}| + \frac{2\bar{d}}{|\bar{\gamma}|} \right) \quad (5)$$

In this expression we must differentiate between 2 possible regimes: the loss-dominated and the avalanche-dominated.

1. In the loss-dominated regime $\bar{\nu} > 1$ ($\bar{\gamma} < 0$), therefore, the steady state solution becomes

$$\bar{n}_\infty \sim \frac{\bar{d}}{|\bar{\gamma}|} = -\frac{\bar{d}}{\bar{\gamma}} \quad (6)$$

In this regime, the runaway electrons are mainly produced by the external source term $\bar{d}$, while the losses limit the population. As a result, the steady-state density is rather small and determined by the balance between the external source and the losses. It is important to note that if this $\bar{d}$ term is 0 then in the stationary regime there will be no runaway electron density.

2. On the other hand, in the avalanche-dominated regime $\bar{\nu} < 1$ ($\bar{\gamma} > 0$), therefore, the steady state solution becomes

$$\bar{n}_\infty \sim \bar{\gamma} + \frac{\bar{d}}{\bar{\gamma}} \quad (7)$$

The main difference we can observe in this new expression is that the steady-state runaway electron density will always be larger than in the previous case. For large gamma, heavily avalanche dominated regime, the density will basically be controlled by the avalanche growth term plus a small correction. However, the density cannot grow indefinitely either, since it is limited by the damping $-\bar{n}^2$ term.

2.3 Exact time-dependent solution

To find the exact time-dependent solution of the runaway electron density we need to solve the differential equation (2). First we find the roots of $\bar{n}$

$$r_{1,2} = \frac{-\bar{\gamma} \pm \sqrt{\bar{\gamma}^2 + 4\bar{d}}}{-2} = \frac{-\bar{\gamma} \pm \beta}{-2}$$

Where:

$$\beta = \sqrt{\bar{\gamma}^2 + 4\bar{d}}, \quad (8)$$

Once we have the roots of $\bar{n}$ and using the variables separation technique, we can write the differential equation as

$$\frac{d\bar{n}}{d\bar{t}} = -(\bar{n} - r_1)(\bar{n} - r_2) \quad \Rightarrow \quad \frac{d\bar{n}}{(\bar{n} - r_1)(\bar{n} - r_2)} = d\bar{t}$$

Solving both integrals

$$\int \frac{d\bar{n}}{(\bar{n} - r_1)(\bar{n} - r_2)} = - \int \frac{A}{\bar{n} - r_1} d\bar{n} - \int \frac{B}{\bar{n} - r_2} d\bar{n} = \frac{1}{r_1 - r_2} \ln \left| \frac{\bar{n} - r_2}{\bar{n} - r_1} \right|$$

$$\frac{\bar{n} - r_2}{\bar{n} - r_1} = k e^{(r_1 - r_2)\bar{t}}$$

And solving for $\bar{n}$ in the previous equation we get

$$\bar{n}(\bar{t}) = \frac{r_2 - r_1 k e^{(r_1 - r_2)\bar{t}}}{1 - k e^{(r_1 - r_2)\bar{t}}}$$

Where using the initial conditions of the problem we can find k

$$\bar{n}(0) = \frac{r_2 - k r_1}{1 - k} = 0 \quad \Rightarrow \quad k = \frac{r_2}{r_1}$$

Ending up with the exact time-dependent solution:

$$\boxed{\bar{n}(\bar{t}) = \frac{\frac{\bar{\gamma} + \beta}{2}(1 - e^{-\beta\bar{t}})}{1 - \frac{\bar{\gamma} + \beta}{\bar{\gamma} - \beta}e^{-\beta\bar{t}}}} \quad (9)$$

Which is of the following form

$$\bar{n}(\bar{t}) = \frac{(\beta + \bar{\gamma}) [1 - e^{-\beta\bar{t}}] / 2}{1 + e^{-\beta\bar{t}}(\beta + \bar{\gamma}) / (\beta - \bar{\gamma})}$$

2.4 Limits and asymptotics behaviors

- First of all, we verify that if we look at the limit when $\bar{t}$ goes to infinity we find the stationary state. Which is true as the following reasoning shows.

$$\lim_{t \rightarrow \infty} \bar{n}(t) = \lim_{t \rightarrow \infty} \frac{\frac{\bar{\gamma} + \beta}{2} (1 - e^{-\beta t})}{1 - \frac{\bar{\gamma} + \beta}{\bar{\gamma} - \beta} e^{-\beta t}} = \frac{\bar{\gamma} + \beta}{2} = \frac{\bar{\gamma} + \sqrt{\bar{\gamma}^2 + 4d}}{2} = \bar{n}_\infty$$

- On the other hand, if we focus on the initial behavior of $\bar{n}$, $\bar{t}\beta \ll 1$, and we use these Taylor expansions

$$e^{-\beta\bar{t}} \sim 1 - \beta\bar{t} + \frac{\beta^2\bar{t}^2}{2}$$

$$\frac{1}{1 - (\beta + \bar{\gamma})\bar{t}} \sim 1 + (\beta + \bar{\gamma})\bar{t}$$

We find that the expression simplifies to:

$$\bar{n}(t) \sim \frac{2\bar{d}\beta\bar{t} - \bar{d}\beta^2\bar{t}^2}{2\beta - (\beta + \bar{\gamma})\beta\bar{t}} \sim \bar{d}\bar{t} \left(1 + \frac{\bar{\gamma}}{2}\bar{t}\right) + O(\bar{t}^3) \quad (10)$$

Taking just the leading term, we see that the run-away electron density at the very start has a linear growth, with $\bar{d}$ being its slope.

$$\bar{n}(t) \sim \bar{d}\bar{t}$$

- Finally, we approximate the temporal evolution of the solution for the loss-dominated case $\bar{\nu} > 1$ ($\bar{\gamma} < 0$). Knowing that for $\bar{\gamma} < 0$, $\beta \sim |\bar{\gamma}| + \frac{2\bar{d}}{|\bar{\gamma}|}$ and neglecting all $\frac{\bar{d}}{\gamma^2}$ terms we end up with the next expression.

$$\bar{n}(\bar{t}) \sim \frac{\frac{\bar{d}}{\bar{\gamma}} (1 - e^{-\beta t})}{1 - \frac{\bar{d}}{\bar{\gamma}^2} e^{-\beta t}} \sim \frac{\bar{d}}{|\bar{\gamma}|} (1 - e^{-\beta t}) \quad (11)$$

3 Numerical method and implementation

3.1 Unified Euler scheme

To solve the normalized equation

$$\frac{d\bar{n}}{d\bar{t}} = f(\bar{n}) = \bar{d} + \bar{\gamma}\bar{n} - \bar{n}^2, \quad (12)$$

we use the unified Euler discretization introduced in the statement:

$$\bar{n}_{k+1} = \bar{n}_k + [\alpha f(\bar{n}_k) + (1 - \alpha) f(\bar{n}_{k+1})] \Delta\bar{t}, \quad (13)$$

where $\Delta\bar{t}$ is the time step and α determines the chosen scheme. In particular,

$\alpha = 1$ corresponds to explicit Euler,

$\alpha = 0$ corresponds to implicit Euler,

$\alpha = 0.5$ corresponds to the semi-implicit scheme.

For the explicit Euler method, the new value $\bar{n}_{k+1}$ is obtained directly from the known value at step k . For the implicit and semi-implicit schemes, the update is nonlinear since $f(\bar{n}_{k+1})$ depends on the unknown next-step value, and an iterative procedure is therefore required.

3.2 Time grid and outputs

The simulations are performed on a uniform time grid. Once the final time $\bar{t}_{\text{final}}$ and the number of time steps `nsteps` are fixed, the time step is given by

$$\Delta\bar{t} = \frac{\bar{t}_{\text{final}}}{\text{nsteps}}. \quad (14)$$

Starting from the initial condition $\bar{n}(0)$, the solution is advanced step by step up to the final time. At each sampled step, the code writes the current time and numerical density to the output file in the form

$$(\bar{t}_k, \bar{n}_k).$$

The last line of the output file also contains the total number of evaluations of the function f , which is useful when discussing efficiency.

3.3 Iterative treatment of the implicit and semi-implicit schemes

For the implicit and semi-implicit methods, the nonlinear update equation is solved iteratively. At each time step, the previous solution is used as an initial guess, and the value of $\bar{n}_{k+1}$ is updated until a convergence criterion is satisfied. More precisely, the iteration stops when the relative difference between two successive approximations becomes smaller than a prescribed tolerance `tol`, or when the maximum number of iterations `maxit` is reached. In the implementation used here, a minimum number of iterations is also enforced in order to stabilize the iterative procedure for the implicit schemes. This approach makes it possible to compare the three Euler methods within the same implementation framework and to study their respective accuracy, convergence, and stability properties.

4 Simulations and analyses

4.1 (a) Case $\bar{\gamma} > 0$: convergence of $\bar{n}(\bar{t}_{\text{final}})$ and $\bar{\tau}$

We now consider the avalanche-dominated regime introduced in the statement, with

$$\bar{t}_{\text{final}} = 32, \quad \bar{n}(0) = 0, \quad \bar{\gamma} = 0.5, \quad \bar{d} = 0.01.$$

For each of the three Euler schemes presented in Section 3.1, simulations were performed for several values of `nsteps`, corresponding to decreasing values of $\Delta\bar{t}$. The numerical results were then compared with the analytical solution derived in Section 2.

Convergence of the final density. The first quantity of interest is the final runaway density $\bar{n}(\bar{t}_{\text{final}})$. For each run, the numerical value at the final time was compared to the exact analytical reference, and the relative error

$$\varepsilon_{n_f} = \frac{|\bar{n}_{\text{num}}(\bar{t}_{\text{final}}) - \bar{n}_{\text{ana}}(\bar{t}_{\text{final}})|}{|\bar{n}_{\text{ana}}(\bar{t}_{\text{final}})|}$$

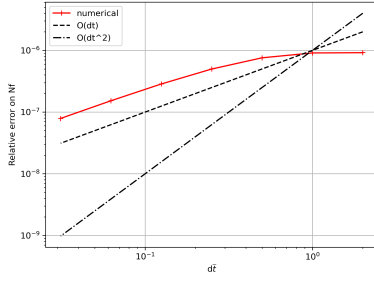
was evaluated. The results are shown in Figure 1. All three methods converge toward the analytical solution as the time step is reduced. The implicit Euler scheme shows the expected first-order behaviour once the coarsest time step is excluded, while the semi-implicit method converges significantly faster and is consistent with second-order accuracy. For the explicit scheme, the error on the final density becomes very small already for moderate time resolution, so the slope is less clearly identifiable on the log-log plot. This is consistent with the fact that at $\bar{t}_{\text{final}} = 32$ the solution is already extremely close to its steady state, which makes the final-density observable less sensitive than transient quantities.

Convergence of the characteristic time. The second observable is the characteristic time $\bar{\tau}$ defined by

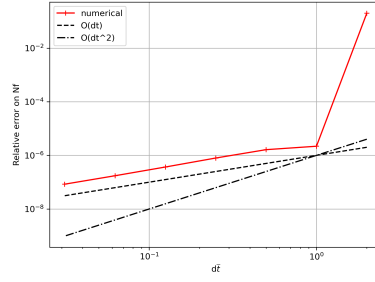
$$\frac{\bar{n}(\bar{\tau})}{\bar{n}_{\infty}} = 0.2.$$

Numerically, $\bar{\tau}$ was obtained by interpolation from the discrete trajectory. The corresponding convergence results are shown in Figure 2. In this case the convergence order is much clearer. The explicit and implicit schemes both behave as first-order methods, whereas the semi-implicit scheme exhibits a much steeper error decay, compatible with second-order convergence. Moreover, for a given time step, the semi-implicit method provides the most accurate estimate of $\bar{\tau}$. Compared with the final density, the characteristic time is more sensitive to time discretization, since it depends on the detailed transient evolution and on the interpolation procedure.

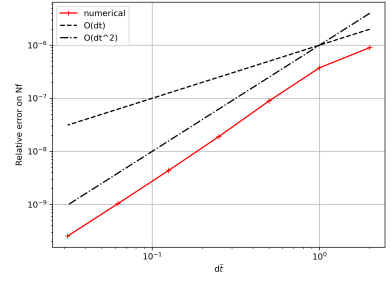
Discussion. Overall, the results are consistent with the theoretical expectations for the three Euler schemes. The explicit and implicit methods display first-order convergence, while the semi-implicit scheme is substantially more accurate and behaves as a second-order method for both observables, especially for the characteristic time. For the present problem, the semi-implicit scheme therefore provides the best accuracy for a given time step.



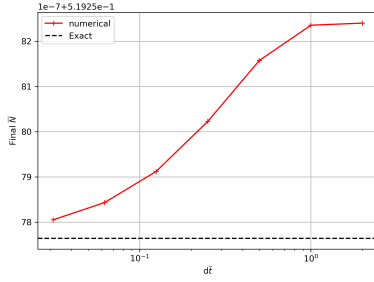
(a) Explicit



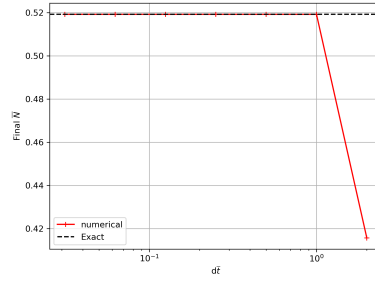
(b) Implicit



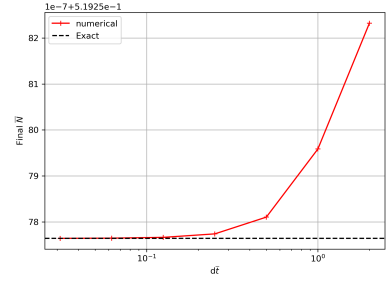
(c) Semi-implicit



(d) Explicit

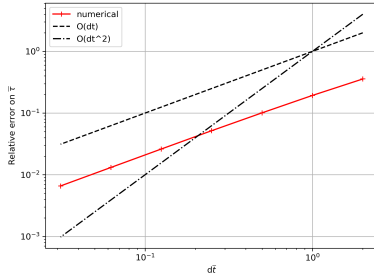


(e) Implicit

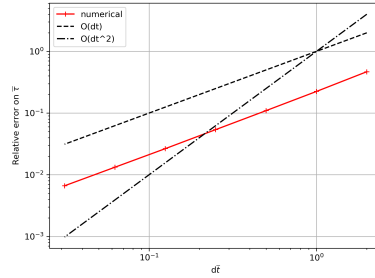


(f) Semi-implicit

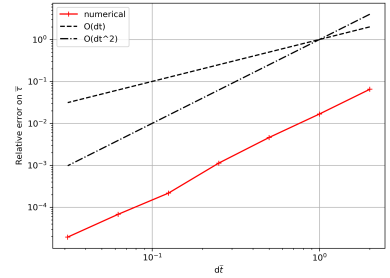
Figure 1: Top row: relative error on the final density as a function of $\Delta \bar{t}$. Bottom row: convergence of the numerical final density toward the analytical reference, for the explicit, implicit and semi-implicit Euler schemes.



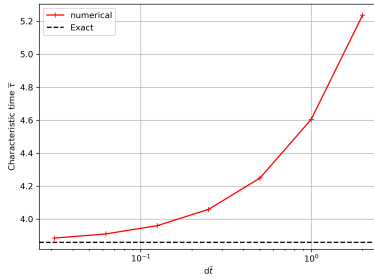
(a) Explicit



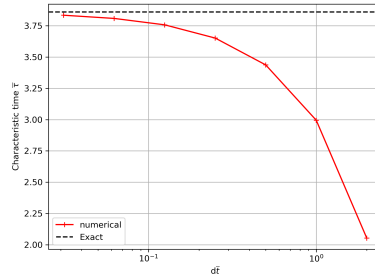
(b) Implicit



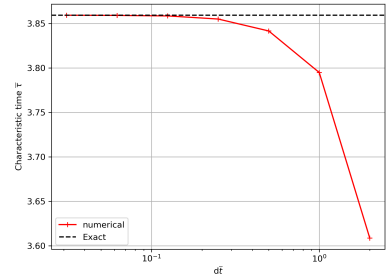
(c) Semi-implicit



(d) Explicit



(e) Implicit



(f) Semi-implicit

Figure 2: Top row: relative error on the characteristic time $\bar{\tau}$ as a function of $\Delta \bar{t}$. Bottom row: numerical characteristic time compared with the analytical reference, for the explicit, implicit and semi-implicit Euler schemes.

4.2 (b) Early-time behaviour: $\beta\bar{t} \ll 1$

In order to study the initial evolution of the runaway density, a high-resolution simulation was performed with

$$\text{nsteps} = 1024, \quad \Delta\bar{t} = \frac{\bar{t}_{\text{final}}}{1024} = \frac{32}{1024} = 0.03125,$$

using the same parameters as in part (a),

$$\bar{n}(0) = 0, \quad \bar{\gamma} = 0.5, \quad \bar{d} = 0.01.$$

A semi-implicit Euler scheme was used in this case.

According to the analytical developments of Section 2.4, the exact solution can be expanded for $\beta\bar{t} \ll 1$ as

$$\bar{n}(\bar{t}) \sim \bar{d}\bar{t} \left(1 + \frac{\bar{\gamma}}{2}\bar{t}\right) + O(\bar{t}^3). \quad (15)$$

In particular, the leading-order behaviour is simply

$$\bar{n}(\bar{t}) \sim \bar{d}\bar{t}, \quad (16)$$

which predicts an initially linear growth with slope $\bar{d}$. Figure 3 shows a zoom on the early-time numerical solution together with the exact analytical curve. The numerical and analytical curves are almost indistinguishable on the plotted interval, which is consistent with the use of a very fine time step and with the smooth, nearly linear behaviour of the solution in the regime $\beta\bar{t} \ll 1$. The figure also confirms that the initial growth is approximately linear, in agreement with the asymptotic expansion derived analytically. This comparison validates both the implementation of the numerical method and the analytical small-time approximation. Using a fine temporal resolution is essential here, since the initial growth takes place on a short timescale and would be less accurately captured with a coarser discretization.

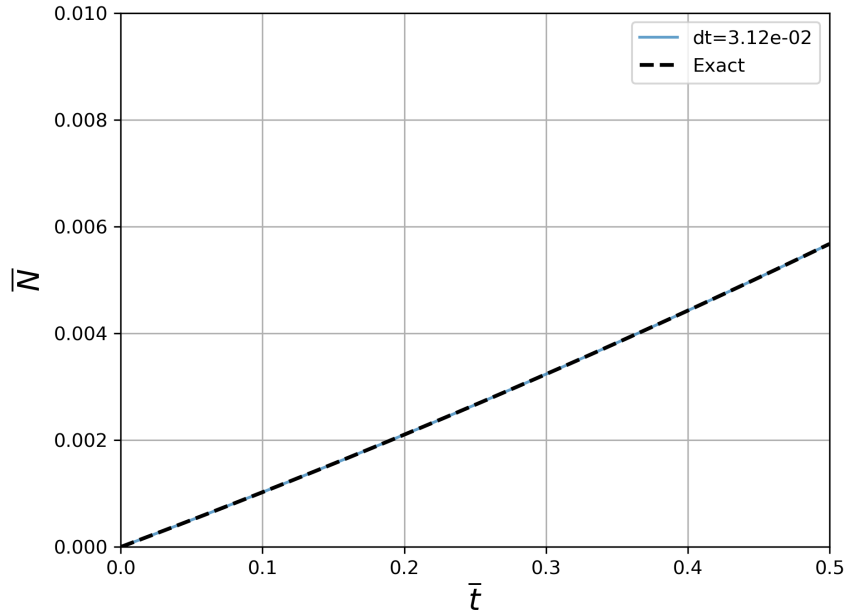


Figure 3: Zoom on the early-time evolution for $\text{nsteps} = 1024$ using the semi-implicit Euler scheme. The numerical and exact analytical curves overlap almost perfectly, confirming the expected behaviour in the regime $\beta\bar{t} \ll 1$.

4.3 (c) Case $\bar{\gamma} < 0$: loss-dominated regime

We now consider the loss-dominated regime by setting

$$\bar{\gamma} = -0.2, \quad \bar{d} = 0.01, \quad \bar{n}(0) = 0, \quad \bar{t}_{\text{final}} = 32.$$

In this regime, the analytical approximation derived in Section 2 predicts a small steady-state value of order

$$\bar{n}_{\infty} \sim \frac{\bar{d}}{|\bar{\gamma}|} = \frac{0.01}{0.2} = 0.05.$$

Using the exact steady-state formula, one obtains

$$\bar{n}_{\infty} = \frac{\bar{\gamma} + \sqrt{\bar{\gamma}^2 + 4\bar{d}}}{2} = \frac{-0.2 + \sqrt{0.08}}{2} \approx 0.04142. \quad (17)$$

Therefore, the analytical estimate is valid at the level of the order of magnitude and captures the physical behaviour correctly, although it slightly overestimates the exact equilibrium value. Figure 4 compares the time evolution obtained with the explicit, implicit and semi-implicit Euler schemes. In all cases, the runaway density starts from zero, increases monotonically, and saturates at a small value. This is consistent with the expected physical picture for $\bar{\gamma} < 0$: the external source term creates a finite runaway population, but losses dominate over avalanche multiplication and prevent the density from reaching the much larger values observed in the avalanche-dominated regime.

The three numerical schemes reproduce this qualitative behaviour correctly. For coarse time steps, the explicit Euler scheme tends to slightly overestimate the final density, whereas the implicit Euler scheme slightly underestimates it. The semi-implicit scheme remains the closest to the analytical reference over the considered range of time steps. As the time step decreases, all three methods converge toward the same asymptotic value. No visible non-physical artefacts such as negativity or oscillations are observed in the present simulations. Hence, in this loss-dominated case, the main differences between the numerical schemes are quantitative rather than qualitative. The explicit and implicit schemes introduce opposite biases for coarse discretizations, while the semi-implicit scheme provides the most accurate overall approximation.

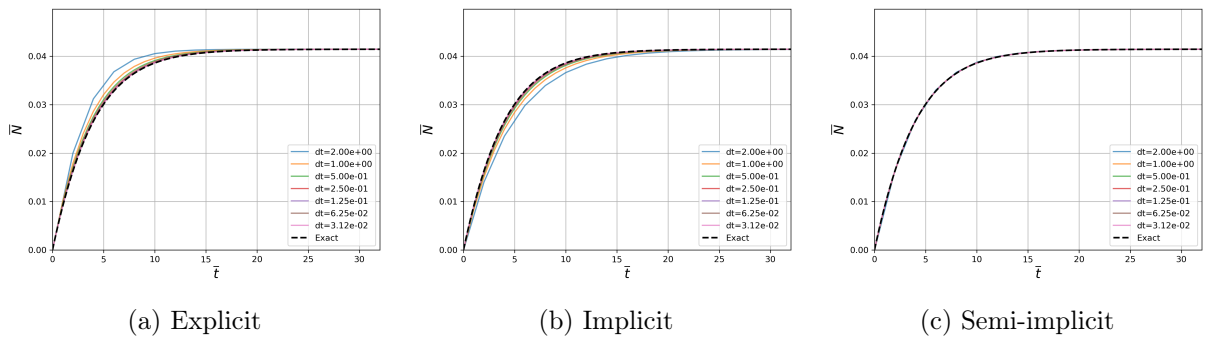


Figure 4: Time evolution of the runaway density in the loss-dominated regime ($\bar{\gamma} = -0.2$) for the three Euler schemes. All methods converge toward a small steady-state value, in agreement with the analytical prediction.

4.4 (d) Stability study (explicit Euler)

We now return to the setting of 4.1 and run explicit Euler with fewer iterations, `nsteps` = 16, 8, 4. It is important to note that when the number of steps is reduced from 16 to 8 and 4, the

time step dt increases, since we have defined it as:

$$d\bar{t} = \frac{\bar{t}_{final}}{n_{steps}}$$

If dt increases, we observe in Fig. 5 that the explicit Euler solution becomes progressively less accurate: for 16 steps the solution is still acceptable, for 8 steps it deviates significantly from the exact one, indeed it starts oscillating; and for 4 steps the behaviour becomes clearly non-physical, with a large deviation and even a negative final value of N . As the time step increases, the solution loses stability, becoming unstable as in the 4 steps scenario, and does not approximate the exact solution anymore.

The behaviour can also be observed in Fig. 6. When the number of time steps is reduced from 16 to 8 and 4, the time step dt increases, the explicit Euler solution strongly deviates from the exact one and may even produce non-physical values.

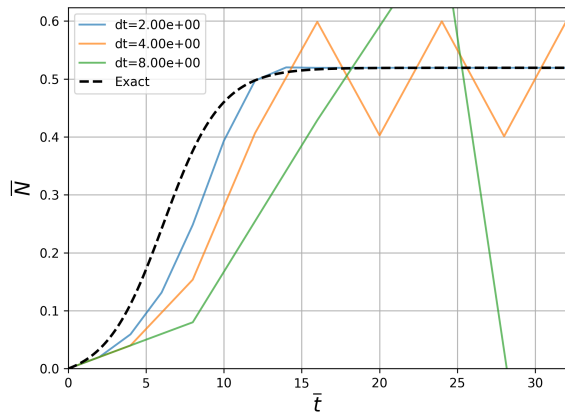


Figure 5: Evolution of $\bar{N}$ for different time steps using the explicit Euler scheme.

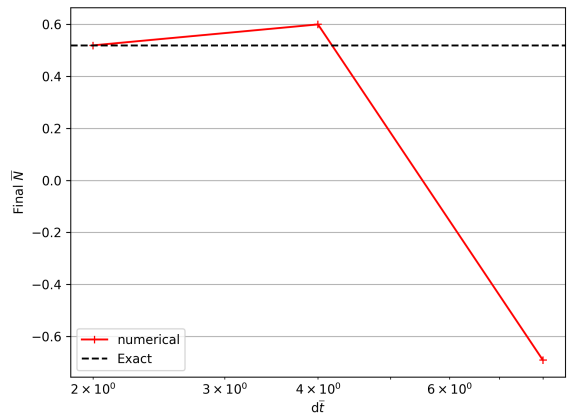


Figure 6: Final value N_f as a function of the time step for the explicit Euler scheme.

This behaviour is consistent with the Von Neumann stability analysis. For a linear mode written as

$$e_{n+1} = e_n[1 + \Delta t \frac{\partial f}{\partial y}(y_n, t_n)] = e_n[1 - \gamma \Delta t]$$

Where for the Euler explicit $\gamma = 1$ Stability requires that the Δt term is not very large, otherwise the solution will be unstable. In fact:

- Cas $0 < \gamma \Delta t < 1$: e_n stable and constant sign.
- Cas $1 < \gamma \Delta t < 2$: e_n stable and alternate sign.
- Cas $\gamma \Delta t > 2$: e_n unstable

The stability condition will be then:

$$|1 - \gamma \Delta t| \leq 1$$

Since we are in the Euler explicit scheme, $\gamma = 1$. Therefore, if dt is too large, this condition is violated and numerical errors grow from one step to the next, leading to the observed instability. We see then that we could have predicted the observed results beforehand with this analysis. For $dt = 2$ it is stable since it is in the mentioned range, however $dt = 4$ and $dt = 8$ will not be stable since the γdt factor is way larger than 2. This explains why the behaviour deteriorates when the number of steps is reduced to 8 and especially to 4.

4.5 (e) (Semi-)implicit scheme without iteration (engine2.cpp)

For the implicit and semi-implicit schemes there is an analytical solution to the discrete equation. We consider the normalized ordinary differential equation (2). Which can also be written as

$$f(N) = d + gN - N^2$$

The generalized Euler scheme used in the code is

$$N_{i+1} = N_i + \Delta t [\alpha f(N_i) + (1 - \alpha)f(N_{i+1})] \quad (18)$$

Substituting the expression of $f(N)$ into the former equation gives

$$N_{i+1} = N_i + \Delta t [\alpha(d + gN_i - N_i^2) + (1 - \alpha)(d + gN_{i+1} - N_{i+1}^2)] \quad (19)$$

Let

$$x = N_{i+1}.$$

Expanding the terms and rearranging them, the equation becomes

$$(1 - \alpha)\Delta t x^2 + [1 - (1 - \alpha)\Delta t g] x - [N_i + \Delta t \alpha(d + gN_i - N_i^2) + \Delta t(1 - \alpha)d] = 0.$$

The solution to this equation is obtained by the quadratic formula

$$N_{i+1} = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}. \quad (20)$$

with

$$A = (1 - \alpha)\Delta t \quad (21)$$

$$B = 1 - (1 - \alpha)\Delta t g \quad (22)$$

$$C = -[N_i + \Delta t \alpha(d + gN_i - N_i^2) + \Delta t(1 - \alpha)d]. \quad (23)$$

Among the two possible roots, the physically relevant solution is chosen, typically the positive or if both of them are, the one closest to N_i .

Using this analytical solution to the discrete equation is a totally different approach as we were doing it before and that has its advantages. For instance, this analytical solution removes the need for an iterative solver for the implicit scheme, thus, reducing the computational cost while providing the same numerical result. In our simulations for example, the iterative approach did 152 steps whereas the analytical one did just 32 and both of them gave practically the same results, specially for low tolerances. This is a considerable decrease of the computational cost.

Moreover, after several simulations with the 4.1 setup and 32 nsteps, it can be observed in Fig. 7 and in Fig. 8 that when you reduce the tolerance of the iterative approach, the $\bar{\tau}$ converges to the value of the analytical solution.

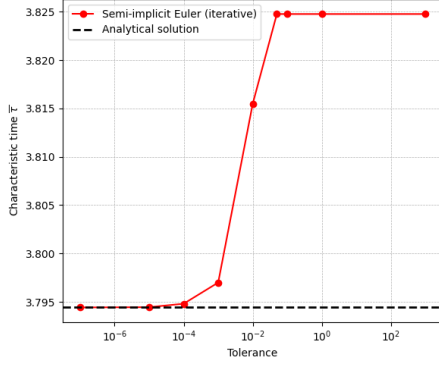


Figure 7: Evolution of $\bar{\tau}$ for different tolerance values using the semi-implicit Euler scheme.

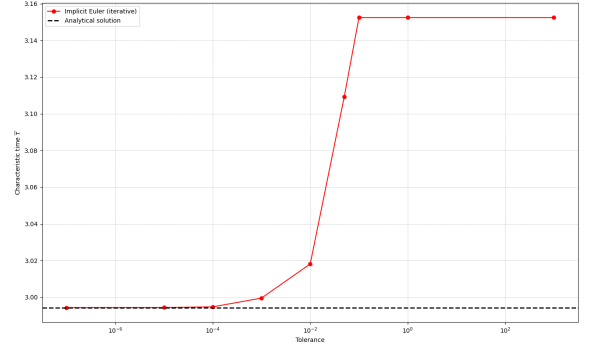


Figure 8: Evolution of $\bar{\tau}$ for different tolerance values using the implicit Euler scheme.

4.6 (f) Optional: efficiency for a target accuracy

- (i) In the original implementation of the code, the iterative solver used for the implicit and semi-implicit schemes includes a constraint enforcing a minimum of three iterations. If we remove this constraint, the solver is no longer obliged to call the function 3 times.

When the time step Δt is significantly reduced, the numerical solution at the next time step is very close to the previous one,

$$N_{i+1} \approx N_i.$$

As a consequence, the function values also remain very similar. This means that the initial guess used in the fixed-point iteration is already very close to the final solution. Therefore the error often becomes smaller than the tolerance after a single iteration, thus ending the iterative loop. As a result, if we increase the tolerance and decrease the time step, the semi-implicit method typically converges in very few iterations. In this regime the semi-implicit scheme behaves almost like an explicit scheme, since it requires only one evaluation of the function `compute_f` per time step.

Consequently, removing the minimum iteration constraint significantly reduces the computational cost while maintaining the same level of accuracy for sufficiently small time steps.

- (ii) Starting from the case 1.3-a, the goal is to determine an optimal combination of the numerical scheme, tolerance, number of forced iterations and time step in order to obtain a relative error on the characteristic time τ smaller than 10^{-4} while minimizing the number of evaluations of `compute_f`. Different configurations were tested by varying the time step Δt , the tolerance of the iterative solver and the number of forced iterations in the implicit and semi-implicit schemes. The efficiency of each configuration was evaluated using the total number of calls to `compute_f`, which represents the computational cost.

The explicit scheme requires only one evaluation of `compute_f` per time step, but achieving the required accuracy on τ generally requires a very small time step. This leads to a large number of time steps and therefore a relatively high computational cost. The implicit scheme is more stable and allows larger time steps, but each time step requires several iterations of the fixed-point solver, increasing the number of evaluations of `compute_f`.

The semi-implicit scheme provides a good compromise between stability and computational cost. For sufficiently small time steps and a moderate tolerance (for example `tol` = 10^{-4}),

the method typically converges in only one or two iterations. In this regime, the semi-implicit method achieves the required accuracy on τ while keeping the number of evaluations of `compute_f` relatively low. Therefore, the optimal configuration is obtained using the semi-implicit scheme ($\alpha = 0.5$), a moderately small time step, a tolerance of order 10^{-4} , and removing the minimum number of forced iterations as we did before. This configuration achieves a relative error on τ below 10^{-4} with the smallest number of evaluations of `compute_f`.

5 Conclusion

In this work, a simplified 0D model for the runaway-electron density was investigated analytically and numerically. The analytical study provided the steady-state solution, the exact time-dependent solution, and useful asymptotic descriptions of both the early-time regime and the loss-dominated regime.

The numerical results confirm that all three Euler schemes reproduce the analytical solution when the time step is sufficiently small. The explicit and implicit methods exhibit first-order convergence, whereas the semi-implicit scheme is more accurate and behaves consistently with second-order convergence, especially for the characteristic time $\bar{\tau}$.

The study of the regime $\beta\bar{t} \ll 1$ confirmed the analytical small-time expansion, with an initial growth that is well approximated by the linear law $\bar{n}(\bar{t}) \sim \bar{d}\bar{t}$. In the loss-dominated case, the runaway density remains small and saturates at a finite value, in agreement with the analytical estimate. Finally, the stability study of the explicit Euler method showed that coarse time steps may lead to strong numerical deviations and even non-physical behaviour.

Overall, among the schemes studied here, the semi-implicit Euler method provides the best compromise between accuracy and robustness.

References

- [1] Exercise statement for *Physique Numérique — Exercice 1*, on the analytical and numerical study of a 0D runaway-electron model, including explicit, implicit and semi-implicit Euler schemes.
- [2] Lecture notes and course material of *Physique Numérique*, Prof. Laurent Villard.
- [3] Use of large language model ChatGPT for language revision, LaTeX editing assistance, and code debugging support.